

Chapter 34 — Perinatal Urology

- **Urinary tract dilation (UTD) is the second commonest prenatal anomaly** after cardiac defects, found in **1–3% of pregnancies**.
- **Most UTD is detected in the second trimester** → parents can be counselled well before delivery.
- **Most causes never need surgery** — transient dilation, ureteropelvic junction (UPJ) pattern of dilation, vesicoureteral reflux (VUR) — but a subset is at real risk of urological disease.
- Hence the need for **explicit thresholds** for prophylactic antibiotics, lower urinary tract imaging and more invasive upper tract imaging.
- Scope: grading systems, prenatal consultation and intervention, imaging modalities and indications, differential diagnosis and outcomes, postnatal risk stratification, and perinatal urological emergencies.
- *Videos 34.1–34.3.* Prenatal UTD of the fetal kidney; measurement of the amniotic fluid index (AFI); fetal ultrasound of a multicystic dysplastic kidney (MCDK).

PRENATAL IMAGING

Maternal Fetal Ultrasound

- **Metanephric kidney appears at day 28 of gestation; urine production begins in earnest at 14 weeks** and can be seen on prenatal ultrasound (US).
- **American College of Obstetricians and Gynecologists (ACOG): first screening US at 18–20 weeks.** In the United States it is done at about 20 weeks' gestational age (GA). **Low-risk pregnancies average two scans; high-risk pregnancies average four.** Exact timing is left to the physician.
- **Referral to maternal–fetal medicine:** there are no set criteria, but **increasing UTD and/or an abnormal AFI** should prompt it.
- **AFI = sum of the four largest pockets of amniotic fluid** (Fig. 34.1). **Normal 8–18 cm; oligohydramnios <5–6 cm; polyhydramnios 20–24 cm and above.**
- **US detects dilation well** (sensitivity up to about 90%), but scan numbers and costs keep rising **without evidence that prenatal detection improves outcome.**
- **How well prenatal US predicts the postnatal diagnosis depends on the condition:**
 - **Severity of prenatal UTD rises linearly with UPJ obstruction**, but not with VUR or duplex anomalies.
 - **Bladder/cloacal exstrophy and neural tube defects** are diagnosed correctly in most cases. **Disorders causing genital ambiguity** are diagnosed much less consistently.
- **Main benefit of prenatal detection: dialogue with the family** on differential, intervention, postnatal imaging and natural history.

Fetal Magnetic Resonance Imaging

- **Fetal magnetic resonance imaging (MRI) has a limited role in UTD.** It is indicated when **specific anatomical detail** is needed, usually in rare conditions: cloacal anomaly, urogenital sinus, cloacal/bladder exstrophy, and duplication anomalies causing bladder outlet obstruction (Fig. 34.2, fetal MRI of a cloacal anomaly).
- **MRI gives better anatomical detail than US**, which can be limited by **amniotic fluid volume, maternal habitus, fetal position and the pelvic bones.**

KEY POINTS

- First maternal screening US at **about 20 weeks**; low-risk pregnancies average **two scans**, high-risk **four**.

- UTD is the **second commonest prenatal finding** and occurs in **1–3%** of pregnancies.
- US does not reliably predict severity for all conditions. It **correlates best with obstructive lesions** (UPJ obstruction, posterior urethral valves [PUV]) and **least with VUR**.
- Prenatal detection lets the family **meet a specialist before delivery** and learn the expected course.

Concept — Prenatal US is a **sensitive detector of dilation but a poor predictor of diagnosis**. Its value is in triage and counselling, not in proving benefit to outcome.

GRADING SYSTEMS

- **There is no consensus on the best grading system.** A multispecialty survey (obstetricians, paediatricians, paediatric urologists, paediatric radiologists) found **anterior–posterior diameter (APD) used mainly prenatally**, and **Society for Fetal Urology (SFU) or subjective (mild/moderate/severe) grading used mainly postnatally**.
- The three main systems are the **anterior–posterior renal pelvic diameter (APRPD)**, **SFU** and **UTD** systems. They are objective, subjective, or a combination of both.

Anterior-Posterior Renal Pelvic Diameter

- **Most common prenatal system** among maternal–fetal specialists. It is **strictly objective** and simple.
- **Technique.** Diameter of the renal pelvis on a **transverse image at the mid-kidney**, using the **renal hilum** as the landmark.
- **Drawback: operator dependence.** The classic error is **measuring in the sagittal instead of the transverse plane**.
- **Physiological APD: <4 mm before 28 weeks; <7 mm at ≥28 weeks.**
- **APD correlates positively with GA**, probably because physiological urine output rises in the third trimester.
- **Severity of prenatal dilation rises linearly with obstructive lesions** (UPJ obstruction, PUV) but **relates inconsistently to VUR** (Table 34.2).
- Several centres report an **APD threshold that predicts UPJ surgery**.
- **Great Ormond Street randomised trial.** Patients with third-trimester APD ≥15 mm and differential renal function (DRF) >40% were randomised to pyeloplasty or observation.
 - **Surgery** → function preserved and dilation reduced in almost all.
 - **Observation** → about half improved and a third were stable. A minority **lost function and needed surgery; their APD was 20–40 mm**.
 - There are **no long-term data**, so only short-term conclusions can be drawn.
- **Threshold trade-off: low cut-off → sensitive but over-tests; high cut-off → specific but misses disease and risks function.**

TABLE 34.1 CLASSIFICATION SYSTEM FOR PRENATAL ANTERIOR-POSTERIOR DIAMETER SEVERITY

Degree	APD, second trimester	APD, third trimester	Proportion of prenatal UTD (%)
Mild	4 to <7 mm	4 to <9 mm	56.7–88.0
Moderate	7 to ≤10 mm	9 to ≤15 mm	10.2–29.8
Severe	>10 mm	>15 mm	1.5–13.4

Data from Feldman et al., 2001; Ahmad and Green, 2005; Wollenberg et al., 2005; Lee et al., 2006; Coelho et al., 2007, 2008.

- **Purpose:** three tiers with **higher third-trimester cut-offs**, as APD grows with GA.
- **Pattern:** frequency is **inversely related to severity**. Mild dilation makes up most cases; severe dilation is rare.
- **Trap:** 10 mm is "severe" in the second trimester but only "moderate" in the third. **A value means nothing without its GA.**

TABLE 34.2 INCIDENCE OF DISEASE FOR SEVERITY OF ILLNESS WITH PRENATAL ANTERIOR-POSTERIOR DIAMETER

Postnatal pathology (% of group)	Mild	Mild-moderate	Moderate	Moderate-severe	Severe	Trend P
Any pathology	11.9	39.0	45.1	72.1	88.3	<.001
UPJ obstruction	4.9	13.6	17.0	36.9	54.3	<.001
Vesicoureteral reflux	4.4	10.8	14.0	12.3	8.5	.10 (no trend)
Posterior urethral valves	0.2	0.9	0.9	6.7	5.3	<.001
Ureteral obstruction	1.2	11.7	9.8	10.6	5.3	.025
Other*	1.2	1.9	3.4	5.6	14.9	.002

Meta-analysis; pooled group sizes 587, 213, 235, 179 and 94 (mild → severe); 95% confidence intervals omitted. *Prune-belly syndrome, VATER (vertebral, anal, tracheo-oesophageal, renal) syndrome, solitary kidney, renal mass, unclassified. Modified from Lee RS, Cendron M, Kinnammon DD, Nguyen HT. *Pediatrics* 2006;118:590.

- **Purpose:** prenatal grade → **postnatal probability of disease**.
- **Pattern:** any pathology rises from about 1 in 9 (mild) to nearly 9 in 10 (severe). UPJ obstruction and PUV rise with grade.
- **Trap:** VUR does not trend with grade (P = .10). Even mild dilation carries a VUR risk, and severe dilation does not predict it.

Society for Fetal Urology Grading System

- **Subjective five-point system (grades 0–4)** based on **pelvicalyceal dilation plus integrity of the renal parenchyma**. It is **applied only postnatally** (APRPD is used prenatally).
- *Figure 34.3 — SFU grades* (modified from Timberlake and Herndon, 2013):
 - **Grade 1** — urine barely splits the renal sinus.
 - **Grade 2** — full pelvis; major calyces dilated.
 - **Grade 3** — minor calyces uniformly dilated; **parenchyma spared**.
 - **Grade 4** — **parenchymal compromise** (thinning).
- **Weakness:** intra- and inter-rater variability from subjectivity. **Computer-enhanced visual learning** teaches the system well and reduces that variability.
- **SFU grade predicts renal function and the need for surgery:**
 - **Grade inversely correlates with preserved DRF (>40%). All SFU 3 kidneys retained function; about 70% of SFU 4 kidneys did.** Some use this to **limit renal scintigraphy**.
 - **Early surgery → mostly SFU 4; continued observation → mostly SFU 3.** In delayed surgery grade correlated less well (still mostly SFU 4).
- **Limits:** not used by other specialties; ignores ureter and bladder.

Urinary Tract Dilation Grading System

- Developed by a **multidisciplinary consensus** (obstetrics, radiology, urology) to give **one system across specialties and across the prenatal/postnatal divide**. The term "**UTD**" was chosen to include the whole urinary tract.
- It **standardises terminology, US technique and data reporting**.
 - **Prenatal APD**: fetal spine at 6 or 12 o'clock.
 - **Postnatal APD**: transverse plane at the mid-kidney (hilar vessels); maximal pelvic diameter within the confines of the renal cortex. Serial studies must use the **same lie (supine vs prone)**.
- A **six-parameter template combines APD (objective) and SFU (subjective)** (Fig. 34.4).
- **Postnatal normal**: APD <10 mm with no calyceal or ureteral dilation = transient, clinically insignificant pelvic dilation. It was set to curb postnatal over-testing and parallels the prenatal norms, giving a **stepwise rise in non-pathological APD**: <4 → <7 → <10 mm (Fig. 34.5).

FIGURE 34.4 — STANDARDISED SIX US PARAMETERS OF THE UTD CLASSIFICATION

Parameter	Recorded as	Note
APRPD	mm	Transverse image, maximal diameter of the intrarenal pelvis
Calyceal dilation — central (major) / peripheral (minor)	Yes / No	Central vs peripheral is what separates low from higher risk
Parenchymal thickness	Normal / Abnormal	Subjective assessment
Parenchymal appearance	Normal / Abnormal	Echogenicity, corticomedullary differentiation, cortical cysts
Ureter	Normal / Abnormal	Dilation is abnormal; transient visualisation is normal postnatally
Bladder	Normal / Abnormal	Wall thickness, ureterocele, dilated posterior urethra

Modified from Nguyen HT, Benson CB, Bromley B, et al. J Pediatr Urol 2014;10:992–998, Table 2.

FIGURE 34.5 — NORMATIVE (NORMAL) THRESHOLDS FOR PRENATAL AND POSTNATAL APRPD

Finding	16–27 weeks	≥28 weeks	Postnatal (>48 h)
APRPD	<4 mm	<7 mm	<10 mm
Calyceal dilation (central, peripheral)	No	No	No
Parenchymal thickness and appearance	Normal	Normal	Normal
Ureters, bladder	Normal	Normal	Normal
Unexplained oligohydramnios	No	No	Not applicable

Modified from Nguyen et al., J Pediatr Urol 2014, Table 3.

- **Pattern**: the normal APD ceiling **rises by 3 mm per step** (4 → 7 → 10) as urine output grows. Normality also requires all five other parameters to be normal.

Other Applications of Ultrasound

- **Medullary pyramidal thickness (PT)**: an objective measure with **excellent inter-rater reliability**, versus only fair reliability for SFU grading.
- **PT ≤3 mm predicts the need for pyeloplasty** for UPJ obstruction.

- **Other pyeloplasty risk factors:**
 - **PT <3 mm after day 3 of life.**
 - **PT falling between the first and second postnatal US.**
 - **Low PT ratio** (hydronephrotic vs normal contralateral kidney).
- **A model using APD, PT and renal length relative to the contralateral kidney** predicted pyeloplasty in unilateral hydronephrosis as well as a model that included SFU grade.
- **PT <3 mm → higher risk of reduced relative function on renal scan.**

COMPARISON — APRPD VS SFU VS UTD GRADING SYSTEMS

Feature	APRPD	SFU	UTD
Nature	Objective	Subjective	Objective + subjective
Period used	Mainly prenatal	Postnatal only	Prenatal and postnatal
Grades	Mild / moderate / severe (GA-indexed)	0–4	A1, A2–3 (prenatal); P1, P2, P3 (postnatal)
Parts of tract assessed	Renal pelvis only	Pelvis, calyces, parenchyma	Pelvis, calyces, parenchyma, ureter, bladder (+ amniotic fluid prenatally)
Main weakness	Operator dependent (plane error)	Inter-rater variability; ignores ureter/bladder; used by urologists only	Newer; outcome data still accruing
Main strength	Simple; predicts obstruction	Predicts function and surgery	Shared language; built-in risk stratification

KEY POINTS

- The **frequency** of each grade of UTD is **inversely related to its severity**, while the chance of urological disease rises with severity.
- APD shows a **linear relationship with obstructive lesions** (UPJ obstruction, PUV).
- **Third-trimester APD of 15 mm** has reasonable positive predictive value for UPJ surgery.
- **SFU**: five-point grading on calyceal dilation and parenchymal integrity.
- **UTD** combines APD and SFU, **adds the lower urinary tract**, and provides standardised reporting and risk stratification.
- **PT** may reliably identify kidneys at risk from UPJ obstruction.

Concept — Each system **trades objectivity against completeness**. APD is reproducible but sees only the pelvis, and SFU sees the parenchyma but is subjective. UTD exists to **give obstetricians and urologists one language across the birth divide**.

PRENATAL MULTIDISCIPLINARY CONSULTATIONS

- **Conditions that may need immediate or urgent postnatal intervention are best served by a multidisciplinary prenatal consultation**, with an agreed plan and agreed responsibilities:
 - cloacal/bladder exstrophy;
 - cloacal/urogenital sinus malformation;
 - autosomal recessive polycystic kidney disease (ARPKD);
 - **bladder outlet obstruction**: PUV, prune-belly syndrome (PBS), urethral atresia;
 - myelomeningocele/neural tube defect spectrum.

Parental Counseling for Fetal Intervention

- **Team for lower urinary tract obstruction (LUTO):** maternal–fetal obstetrician, paediatric urologist, paediatric nephrologist and neonatologist.
- **Three statements can be made reliably to parents:**
 - **Fetal intervention improves perinatal survival when indicated**, but carries a **complication rate of over 40%** to mother and fetus combined.
 - **Selective treatment does not reliably reduce the long-term need for renal replacement.** That burden may be large and lifelong.
 - **Each case needs an individual risk–benefit analysis** with shared decision-making in a multidisciplinary setting.

Parental Counseling for Prenatal Surgery for Myelomeningocele Closure

- **Team:** maternal–fetal obstetrician, paediatric neurosurgeon, paediatric urologist and neonatologist.
- **Work-up:** complete fetal survey including **echocardiogram**, plus **karyotype**.
- **Parents must understand that fetal closure is experimental.** It **reduces ventriculoperitoneal (VP) shunting and improves cognitive/motor function**, but has **no meaningful effect on bladder function**.
- **ACOG / Society for Maternal-Fetal Medicine recommendations (2017).**
 - Open maternal–fetal repair **improves important paediatric outcomes at the cost of maternal and fetal procedural risk.**
 - Women who meet the criteria for in-utero repair should be counselled **non-directively** about all options, including open fetal surgery.
 - Interested candidates should be **referred to a fetal therapy centre** with the expertise, team and intensive care facilities.

KEY POINTS

- Prenatal intervention for LUTO **improves perinatal survival but does not reliably preserve long-term renal function.**
- Counselling on fetal myelomeningocele closure belongs at a **centre with fetal surgery expertise** and a multidisciplinary team.
- Fetal myelomeningocele surgery **improves motor/cognitive skills and reduces VP shunting**, with **no clear bladder benefit.**

Concept — *Fetal surgery buys what it directly treats: lungs in LUTO and hindbrain/neurological outcome in myelomeningocele. Organs damaged before the operation (kidneys, bladder) are not rescued, so counselling must separate survival from renal and bladder outcome.*

URINARY TRACT DILATION

Risk Stratification and Management for Prenatal Urinary Tract Dilation

- The six UTD parameters assign two prenatal risk levels: **A1 (low)** and **A2–3 (increased)** (Fig. 34.6).
- The **APD threshold steps up at 28 weeks**, allowing for fetal growth and the physiological rise in third-trimester urine output.
- **Prenatal follow-up depends on risk level:**
 - **A1 found before 28 weeks → repeat US after 32 weeks.**

- **A2-3 → US every 4 weeks until delivery.** Two reasons: the lesion may **progress and change the delivery plan/location**, and a **second fetal anomaly is more likely**.
- **Both groups → postnatal US after 48 hours** (Fig. 34.7).

FIGURE 34.6 — RISK STRATIFICATION FOR PRENATAL UTD

Parameter	A1 — low risk	A2-3 — increased risk (any one)
APRPD 16-27 weeks	4 to <7 mm	≥7 mm
APRPD ≥28 weeks	7 to <10 mm	≥10 mm
Calyceal dilation	Central or none	Peripheral
Parenchymal thickness / appearance	Normal	Abnormal
Ureters	Normal	Abnormal
Bladder	Normal	Abnormal
Oligohydramnios	No unexplained oligohydramnios	Unexplained oligohydramnios (presumed genitourinary cause)

Modified from Nguyen et al., J Pediatr Urol 2014, Figure 3. Central and peripheral calyceal dilation may be hard to judge early in gestation.

- **Pattern:** A1 = **isolated pelvic dilation**; any one abnormal parameter → A2-3.
- **Trap:** **A2-3 is one tier** prenatally. Only the postnatal system splits into three (P1-P3).
- *Figure 34.7.* A1: transverse mid-kidney APD <10 mm at 20 weeks. A2-3: peripheral calyceal dilation with parenchymal thinning, APD >10 mm.

Prenatal Management

Prenatal Intervention for Lower Urinary Tract Obstruction

- **Male fetus + severe bilateral UTD (bladder, ureters, kidneys) + reduced amniotic fluid → evaluate for LUTO.** LUTO is a spectrum, so each case is individualised. Counselling weighs fetal benefit against the risk to both fetus and mother.
- **Amniotic fluid changes from placental origin to fetal urine at about 16 weeks.** Restoring fluid aims to allow **pulmonary development**, and possibly renal development.
- **Oligohydramnios presenting after 27 weeks → favourable pulmonary outcome** (observational cohort). So intervention is reserved for oligohydramnios developing **very early in the second trimester**.
- **A recent series challenges this:** vesicoamniotic shunt (VAS) placement **before 16 weeks** gave better pulmonary and renal function than later placement.
- **Need for intervention is rare, about 1 in 60,000.**
- **Systematic reviews: intervention is feasible, but evidence is insufficient that it improves long-term renal function.** Survival improves, partly because fetuses without restored fluid die in utero or as neonates.
- **Options:** early delivery, fetal shunting, fetal surgery (cystoscopy), serial amnioinfusion. Early delivery is favoured more by non-US physicians when bladder outlet obstruction is found in the early third trimester, lungs are mature and fluid is falling.
- **Setting:** operating room, **maternal epidural**. **Fetal anaesthesia** = narcotic and/or paralytic injected into the **umbilical vein under US guidance**. Access is percutaneous for both VAS and fetal cystoscopy.
- **Complications (early series: 44% overall):**
 - incomplete drainage (commonest);
 - premature labour;

- urinary ascites;
- chorioamnionitis;
- case reports of **bowel herniation** and **direct fetal injury** (Fig. 34.8: bowel herniation and umbilical strangulation after VAS).
- **Commonest technique: percutaneous VAS** (Fig. 34.9); **Rodeck shunts lie flush on the abdominal wall** to limit migration, unlike the **Harrison** shunt.
- **Advantage of fetal cystoscopy: direct view of the posterior urethra → diagnosis → directed treatment.**
- *History (1982)*. First clinical series of fetal intervention for bladder outlet obstruction. Both fetuses treated by open fetal surgery died of neonatal renal failure. Of those with equivocal function, three were shunted and two survived and did well.
- *Long-term VAS outcome (Philadelphia)*.
 - **Fetal urine profile predicted early neonatal survival.**
 - Diagnoses: PUV, urethral atresia and PBS.
 - **More than half had renal insufficiency; a third needed renal replacement.**
 - Bladder function was judged adequate (a third needed clean intermittent catheterisation [CIC]; most voided spontaneously), but **no urodynamics were done.**
 - **Quality of life was similar to the normal population** and better than in other chronic diseases.
- *PLUTO (Percutaneous shunting in Lower Urinary Tract Obstruction) — landmark multicentre randomised trial.*
 - **Population:** fetal bladder outlet obstruction where the benefit of intervention was uncertain.
 - **Stopped early for poor recruitment:** 31 randomised (16 VAS, 15 conservative).
 - **Perinatal death was higher with conservative management**, but there was **no meaningful effect on renal function** (Fig. 34.10).
 - **Complications 7/16 (44%):** premature rupture of membranes (3) and non-functioning shunts (4).
- *Figure 34.10 — PLUTO flow (Morris et al., Lancet 2013)*.
 - **VAS arm (16):** 3 terminations/1 intrauterine death → 12 live births → 4 neonatal deaths (pulmonary hypoplasia) → 8 alive at 28 days → 7 alive at 2 years (2 normal function, 5 moderate impairment).
 - **Conservative arm (15):** 2 terminations/1 intrauterine death → 12 live births → 8 neonatal deaths (pulmonary hypoplasia) → 4 alive at 28 days → 3 alive at 2 years (2 moderate impairment, 1 end-stage renal failure).
 - **Crossover** occurred in both arms. The trial is **underpowered**.

Fetal Cystoscopy

- **Same aims as VAS:** decompress, restore amniotic fluid for lung development, possibly help the kidneys.
- **Advantages:**
 - **direct visualisation gives a fetal diagnosis** of the cause of LUTO;
 - **directed treatment** (laser valve ablation) restores **bladder cycling**;
 - **no dependence on a device** staying functional through the pregnancy.
- **Complications:** as for VAS, **plus urethrocuteaneous and rectourethral fistula (up to 10%)**. The cause is **antegrade access at an anatomical disadvantage plus laser energy**.

Vesicoamniotic Shunt Placement Versus Fetal Cystoscopy

- **Multicentre retrospective comparison** of VAS or cystoscopy with laser ablation versus expectant management. Strict inclusion: diagnosis at **16–26 weeks**, severe LUTO, no other anomaly (including normal echocardiogram), **favourable urine electrolytes, normal male karyotype**.
- **All LUTO: both VAS and fetal cystoscopy improved survival** versus observation.
- **Postnatally confirmed PUV: both improved 6-month survival, but only fetal cystoscopy improved renal function**.
- **Newer shunts (Somatex) have a favourable safety profile**, with good 2-year survival. Placement even **before 14 weeks** caused no early membrane rupture, miscarriage or maternal complication.
- **Ruano severity staging, endorsed by the SFU** (Fig. 34.11), is indexed to perinatal and renal outcome to select candidates for intervention. **Stage III** had more interventions, more deaths within 24 h, higher overall mortality and more renal replacement.

FIGURE 34.11 — RUANO CLASSIFICATION OF PRENATAL LOWER URINARY TRACT OBSTRUCTION

Parameter	Stage I — mild LUTO	Stage II — severe LUTO, renal function preserved	Stage III — severe LUTO, renal function abnormal
Amniotic fluid	Normal	Oligo- or anhydramnios	Oligohydramnios, usually anhydramnios
Renal echogenicity	Normal	Hyperechogenic	Hyperechogenic
Renal cortical cysts	Absent	Absent	Can be present
Renal dysplasia	Absent	Absent	Can be present
Fetal urine biochemistry	Favourable	Favourable within three consecutive samples	Not favourable after three consecutive samples
Fetal intervention	Not indicated	Indicated — to prevent pulmonary hypoplasia and severe renal impairment	May be indicated — for pulmonary hypoplasia only, not renal impairment; further study needed

Disease may progress I → II → III during pregnancy. Modified from Farrugia MK, Braun MC, Peters CA, Ruano R, Herndon CD. J Pediatr Urol 2017;13:345–351.

- **Purpose:** pick the fetus in whom **kidneys are still salvageable**, which is stage II.
- **Pattern:** stage II and III have the same fluid and echogenicity. The **discriminators are cortical cysts, dysplasia and urine biochemistry**.
- **Trap:** stage III can still be treated to **save the lungs**, but the kidneys are already lost.

Serial Amnioinfusion

- **Serial amnioinfusion is an emerging way to promote lung development** in bladder outlet obstruction and in **congenital bilateral renal agenesis (CoBRA)**.
- **In CoBRA there is no urine and no bladder filling, so shunting or cystoscopy cannot work.** Case reports are promising.
- **RAFT (Renal Anhydramnios Fetal Therapy) trial:** prospective multi-institutional trial of serial amnioinfusion for anhydramnios, currently under way.
- **No reliable prenatal predictor of postnatal renal function exists.** Fetal urine **sodium, chloride, osmolality and β 2-microglobulin** may predict the need for renal replacement but are not validated.
- **European Reference Network for Rare Kidney Diseases consensus (2022):** amniotic fluid volume, parenchymal echogenicity and fetal urine biomarkers should **not** be used as accurate predictors of postnatal renal function.

COMPARISON — VESICOAMNIOTIC SHUNT VS FETAL CYSTOSCOPY

Feature	Vesicoamniotic shunt	Fetal cystoscopy
Diagnosis of cause	No	Yes — direct view of posterior urethra
Mechanism	Bypass bladder to amniotic space	Laser ablation of valve → bladder cycles
Device dependence	Yes — migration, blockage, dislodgement	No
Specific complications	Non-functioning shunt, bowel herniation, umbilical strangulation	Urethrocuteaneous/rectourethral fistula (up to 10%)
Survival vs observation	Improved	Improved
Renal function (PUV)	Not improved	Short-term improvement; no long-term data

KEY POINTS

- **Predictors of poor outcome:** oligohydramnios before 27 weeks, renal cortical cysts, increased echogenicity.
- Prenatal LUTO intervention improves **perinatal survival**, not long-term renal function.
- Fetal cystoscopy can both **diagnose and treat** LUTO.
- Fetal cystoscopy may give a **short-term renal advantage over VAS**; there are no long-term data.
- **Serial amnioinfusion** may improve lung outcome in anhydramnios and is being tested in a trial.

Concept — Everything in fetal LUTO management follows from one fact: **after 16 weeks amniotic fluid is fetal urine, and without fluid the lungs do not develop**. Intervention reliably treats the lung problem. The kidney problem is often established dysplasia before the needle goes in.

POSTNATAL EVALUATION AND MANAGEMENT

- **Postnatal UTD management has shifted to an evidence-based, risk-stratified model.** Imaging and prophylactic antibiotics are tailored to the **risk of urinary tract infection (UTI), renal deterioration or surgery**. Historically they were driven by the chance of detecting VUR (up to about a quarter of patients).

Imaging**Renal Ultrasound/Resolution of Urinary Tract Dilation**

- **First postnatal US after 48 hours.** Newborns are relatively **intravascularly depleted: third-spaced fluid is mobilised over the first 24–48 h**, so an earlier scan under-reads dilation. Some delay the scan up to **2 weeks** without harm.
- **Follow the consensus standardisation** of technique and reporting.
- **Three postnatal risk levels, P1–P3**, set by APD plus the other five parameters (Fig. 34.12, below). Calyceal or ureteral dilation may be isolated in P2; parenchymal cysts or thinning make P3.
- **SFU Hydronephrosis Registry (prospective, multicentre): SFU 1 → 100% resolved by about 2.5 years; SFU 2 (P1) → about 90–95% resolved by 4 years.** Another prospective registry found SFU and UTD grades equally predictive of resolution.

FIGURE 34.12 — RISK STRATIFICATION FOR POSTNATAL UTD (>48 H)

Parameter	P1 — low risk	P2 — intermediate risk	P3 — high risk
APRPD	10 to <15 mm	≥15 mm	≥15 mm

Parameter	P1 — low risk	P2 — intermediate risk	P3 — high risk
Calyceal dilation	Central	Peripheral	Peripheral
Parenchymal thickness	Normal	Normal	Abnormal
Parenchymal appearance	Normal	Normal	Abnormal
Ureters	Normal	Abnormal	Abnormal
Bladder	Normal	Normal	Abnormal
Approximate SFU equivalent	SFU 1–2	SFU 3	SFU 4

Modified from Nguyen et al., J Pediatr Urol 2014, Figure 6. Classification is by the highest-risk feature present.

- **Pattern:** parenchyma decides P2 vs P3; peripheral calyces or ureter decide P1 vs P2.
- **Trap:** an APD above 15 mm with normal calyces, ureter and parenchyma is not automatically P3. Also, parenchymal thinning makes a kidney P3 at any APD.

Voiding Cystourethrogram

- **Dilation correlates poorly with VUR** (present in up to 31%); up to 25% with VUR have a normal first US.
- **American Academy of Pediatrics (AAP) standardised voiding cystourethrogram (VCUG):**
 - **ALARA** ("as low as reasonably achievable") radiation;
 - **non-balloon catheter**;
 - **bladder cycling** for reliability;
 - **images:** kidney–ureter–bladder (KUB) scout, filling and voiding phases.
- **Target VCUG at UTI risk, not VUR detection:** females, uncircumcised males, high-grade dilation, ureteral dilation, bladder abnormality; shared decision with the family.
- **Outpatient VCUG is acceptable unless bladder outlet obstruction is suspected:** thick-walled bladder (≥ 0.5 cm), distended bladder, bilateral hydroureteronephrosis.
- **Active surveillance of known VUR is safe in stable patients.** The historical benefit of finding asymptomatic VUR is unproven.

Contrast-Enhanced Voiding Urosonography

- **Contrast-enhanced voiding urosonography (CeVUS):** microbubbles instilled by urethral catheter; VUR detected on US. **No radiation.**
- **Possibly more sensitive than VCUG for VUR** (limited data).
- **Poor view of the bladder and urethra** → a formal VCUG is still needed when anatomical detail matters (e.g. the posterior urethra).

Renal Scintigraphy

- **Measures differential function, cortical defects and drainage.** Its weakness is lack of anatomical detail.
- **Technetium-99m mercaptoacetyltriglycine (^{99m}Tc -MAG3) is preferred over diethylenetriaminepentaacetic acid (DTPA) for obstruction or impaired function** because of better extraction.
 - **About 50% of MAG3 is extracted by the proximal tubule per pass** and secreted into the collecting system.
 - Uptake is **linear with renal blood flow**, so it gives relative function against the contralateral kidney.
- **"Well-tempered" diuretic renogram (Conway; SFU / Society of Nuclear Medicine)** standardises hydration, bladder drainage, tracer dose, diuretic dose and timing, and data acquisition and interpretation.

- **Radiation is generally low, but dose rises linearly with degree of obstruction.**
- **MAG3 findings that raise consideration of surgery in an asymptomatic infant (Fig. 34.13):**
 - **DRF <40%;**
 - **T½ >20 minutes;**
 - **significant retention on delayed upright imaging.**

UPJ indications are detailed in Chapter 53.

- **Pitfalls:**
 - **False-positive "obstruction"** from dehydration, or from **furosemide given too early** into a capacious system. The diuretic should be given at **peak tracer uptake**, which varies between patients. A formal **upright protocol** mitigates this.
 - **Invasive:** intravenous line plus urethral catheter.
 - **Region-of-interest placement** changes the drainage curve, especially when pelvis and ureter are both dilated.
- **MAG3's role in postnatal UTD is undefined;** given its pitfalls and US's ability to predict DRF >40%, many use it **selectively, relying on serial renal ultrasound (RUS).**
- **^{99m}Tc-dimercaptosuccinic acid (DMSA): bound to the proximal tubule with minimal secretion → a cortical agent with limited role in UTD. Uses:**
 - **cortical defects** (dysplasia in the absence of infection) in VUR;
 - **confirming absent function in suspected MCDK;**
 - **polar function in duplex systems.**
- *Figure 34.13. MAG3 cortical phase (A) and drainage phase (B).*

Magnetic Resonance Urography

- **Advantages of magnetic resonance urography (MRU):** no radiation; best anatomical detail; also gives differential function, glomerular filtration rate and drainage.
- **Drawbacks:** cost, possible need for sedation or anaesthesia, limited availability.
- **Reserved for operative planning that needs anatomy and/or function:** ectopic ureter, megaureter, **lower-pole crossing vessels causing UPJ obstruction.**

COMPARISON — POSTNATAL IMAGING MODALITIES IN UTD

Modality	Answers	Radiation	Key limitation
Renal/bladder US	Grade, parenchyma, ureter, bladder	None	First scan must wait >48 h
VCUG	VUR and urethral anatomy	Yes (ALARA)	Catheter; target by UTI risk
CeVUS	VUR (possibly more sensitive)	None	Poor urethral imaging
MAG3 renogram	Differential function + drainage	Low (rises with obstruction)	False-positive obstruction; invasive
DMSA	Cortical defects, MCDK non-function, polar function	Low	No drainage information
MRU	Anatomy + function + drainage	None	Cost, sedation, availability

KEY POINTS

- First postnatal US at **48 hours**, after **third-spaced fluid returns** to the intravascular space.

- Three risk levels: **P1 (SFU 1–2) APD 10–15 mm; P2 (SFU 3) APD ≥ 15 mm + calyceal dilation; P3 (SFU 4) APD ≥ 15 mm and/or other abnormal parameters.**
- **APD correlates poorly** with the presence and grade of VUR.
- VCUg decisions rest on **UTI risk**, unless LUTO is suspected.

Concept — Each test is ordered to **answer a question that changes management**. Anatomy comes from US, reflux from VCUG, function and drainage from MAG3. **Detecting disease for its own sake has been abandoned**, and that is why VCUG follows UTI risk rather than prevalence.

MANAGEMENT RECOMMENDATIONS FOR POSTNATAL URINARY TRACT DILATION

P1 Urinary Tract Dilation (Low Risk)

- **Most postnatal UTD is P1 and transient.** Two prospective registries (SFU Hydronephrosis Registry, McMaster) inform UTI risk, prophylactic antibiotic (PA) value and resolution.
- **P1 (SFU 1–2) resolves in most children by 24–48 months** → **second postnatal US at 3–6 months.**
- **Both registries agree: no PA and no VCUG for P1 (SFU 1–2),** as defined by the 2014 consensus statement.

P2 Urinary Tract Dilation (Intermediate Risk)

- **Evidence is weakest for P2.** Surveyed paediatric urologists varied on PA (by sex and circumcision status). Most would order a VCUG, and about half a renal scan.
- **PA and VCUG are at the physician's discretion, except ureteral dilation >7 mm** → **PA recommended**, because of the large UTI risk reduction shown in the Hydronephrosis Registry.
- **VCUG can usually be done as an outpatient**, combined with the RUS at the first postnatal visit.
- **Counselling covers:** expected UTI rate, chance of finding VUR, treatment of VUR, natural history.

P3 (High Risk) Urinary Tract Dilation

- **P3 carries the highest risk of UTI, renal deterioration and surgery. PA, lower urinary tract imaging and renal scintigraphy are recommended,** and most SFU survey respondents support this.
- **Dissenting view:** reflux prevalence is raised across all UTD grades, so VCUG should depend on features suggesting **high-grade reflux (ureteral dilation) or LUTO**, not on P grade alone.
- **Second postnatal US at about 1 month**, combined with the VCUG. Suspected LUTO may justify imaging even for less severe upper tract dilation (e.g. bilateral ureteral dilation).
- **Persistent P3** → **MAG3** (combined with VCUG if not yet done).
- **PA without VUR** → **continue until UPJ surgery or age 1 year**, because **most UTIs occur in the first year.**

COMPARISON — POSTNATAL MANAGEMENT BY UTD RISK LEVEL

Element	P1 (SFU 1–2)	P2 (SFU 3)	P3 (SFU 4)
Repeat US	3–6 months	Early; individualised	1 month
Prophylactic antibiotics	No	Discretion; yes if ureter >7 mm	Yes (to UPJ surgery or 1 year if no VUR)
VCUG	No	Discretion (outpatient)	Yes
MAG3	No	Discretion	Yes, if persistent
Expected course	Resolves by 2–4 years	Variable	Highest risk of UTI, loss of function, surgery

Considerations for Severe Bilateral Urinary Tract Dilation

- The protocols above apply to isolated unilateral UTD.
- Severe bilateral dilation of kidneys, ureters and bladder plus reduced amniotic fluid → bladder outlet obstruction until proven otherwise.
 - Males: PUV, urethral stenosis, PBS.
 - Females: ureterocele, with a cystic bladder lesion.
- Megacystis and bilateral VUR also dilate upper and lower tracts but do not cause oligohydramnios.
- After delivery: decompress the bladder → start antibiotic prophylaxis → evaluate promptly to establish the cause.

KEY POINTS

- P1 UTD resolves completely in most children by 4 years.
- P1 (SFU 1–2): no PA, no VCUG.
- P2 (SFU 3): PA and VCUG at discretion, unless ureter >7 mm → PA.
- P3 (SFU 4): PA, VCUG and MAG3.

Urinary Tract Infection

- UTI occurs in 8–22% of infants with prenatal UTD.
- Degree of kidney dilation predicts UTI. A meta-analysis and several cohorts show a stepwise rise in UTI with grade, and high-grade dilation clearly exceeds low-grade.
- Other predictors in single cohorts: high-grade VUR, ureterovesical junction (UVJ) obstruction, ureteral dilation >11 mm.
- SFU registry: about 90% of UTIs occurred in the first year (Fig. 34.14); risk factors were female sex, high-grade dilation and intact foreskin. McMaster registry added ureteral dilation, VUR and no prophylaxis (see registry comparison below).
- Agreed risk factors: female sex, high-grade dilation (P3, SFU 4), uncircumcised status. Still debated: ureteral dilation, VUR, PA. Confounders are non-uniform VUR screening and case mix.
- Circumcision protects males with prenatal hydronephrosis (PHN): number needed to treat (NNT) = 10 to prevent one UTI. A meta-analysis found protection regardless of cause (odds ratio about 0.3). The AAP supports circumcision of newborn males with urological anomalies.
- Figure 34.14 — UTI-free survival by sex, SFU registry (Zee et al., 2016). Girls' UTI-free survival falls steeply in the first year and keeps falling to 30 months. Boys' survival stays above 95% after about 6 months. The difference is significant ($P < 0.001$).

Prophylactic Antibiotics

- The role of PA in prenatal UTD is not clearly defined. Historically nearly everyone with dilation received PA and a VCUG. The reason was the link with VUR, present in up to 25% even when postnatal dilation had resolved.
- Evidence for benefit:
 - a systematic review found PA reduced UTI in high-grade but not low-grade UTD;
 - a retrospective study found a fivefold reduction in UTI with PA in the ureteral-dilation subset;
 - SFU registry: PA protective when ureteral dilation >7 mm;
 - the McMaster registry showed significant risk reduction.

- **Evidence against:**
 - two **SFU registry reports found no UTI reduction** overall;
 - a **systematic review of 11 studies (>3,900 patients)** found **no reduction in pooled UTI rates** with PA in non-randomised studies (about 8% without vs 10% with).
- **Why evidence is weak: UTI is uncommon (8–19%)** → too few events; populations vary in VUR screening and PA use.
- **Paradigm shift: from blanket PA to PA only where a meaningful risk reduction exists.** Candidates are:
 - higher-grade UTD;
 - uncircumcised males;
 - females;
 - selectively, VUR and/or **ureteral dilation >7 mm**.

Most paediatric urologists support this approach.

PROPHYLACTIC ANTIBIOTIC CHOICE BY AGE (CHOOSE BY AGE AND HEPATIC MATURITY)

Agent	Minimum age	Note
Amoxicillin	Newborn	Safe from birth
First-generation cephalosporin	Newborn	Safe from birth
Nitrofurantoin	≥8 weeks	Avoided in early infancy (immature hepatic metabolism)
Trimethoprim derivatives	≥8 weeks	Avoided in early infancy (immature hepatic metabolism)

- **Trap:** nitrofurantoin and trimethoprim are **not neonatal choices**.

COMPARISON — SFU HYDRONEPHROSIS REGISTRY VS MCMASTER REGISTRY

Feature	SFU Hydronephrosis Registry	McMaster registry
Design	Prospective, multicentre	Prospective
High-grade UTD	Minority (about a quarter)	Majority (over half)
Overall UTI rate	8%	19%
VUR as UTI risk factor	No (nor ureteral dilation)	Yes (febrile UTI)
PA effect	No overall reduction (except ureter >7 mm)	Significant reduction
Shared findings	Female sex, high-grade dilation, intact foreskin → UTI; no PA/VCUG for P1	

- **Why they differ:** different **case mix, VUR screening rates and PA use**, reflecting regional referral patterns and centre bias. **They compare different cohorts rather than truly contradicting each other.**

KEY POINTS

- Degree of dilation **predicts UTI**.
- **UTI risk factors:** female sex, intact foreskin, high-grade dilation, ureteral dilation, VUR.
- **PA recommended** for P2 (SFU 3) with **ureter >7 mm**, and for **P3 (SFU 4)**.

Concept — Because UTI after prenatal UTD is **uncommon**, blanket prophylaxis treats many to protect few. Risk factors (sex, foreskin, grade, ureter) therefore **concentrate the events** into a group where prophylaxis can make a measurable difference.

KEY CONDITIONS

Etiology for Prenatal Urinary Tract Dilation

TABLE 34.3 ETIOLOGY AND INCIDENCE FOR ANTENATAL HYDRONEPHROSIS

Aetiology	Incidence (%)
Transient UTD	50–70
UPJ pattern dilation	10–30
Vesicoureteral reflux	10–40
UVJ pattern dilation	5–15
Multicystic dysplastic kidney	2–5
Duplex system, ureterocele, ectopic ureter	5–7
Lower urinary tract obstruction (PUV, PBS)	0.1–0.5

Modified from Nguyen HT, Herndon CDA, Cooper C, et al. The Society for Fetal Urology consensus statement on the evaluation and management of antenatal hydronephrosis. J Pediatr Urol 2010;6:212–231, Table 5.

- **Purpose:** the **pre-test probabilities** for counselling.
- **Pattern:** **transient dilation is commonest; UPJ obstruction is the commonest surgical cause; LUTO is rarest but most serious.**
- **Trap:** **VUR is as common as UPJ-pattern dilation**, and it is the one cause not predicted by the grade of dilation.

Transient Urinary Tract Dilation

- **Resolution rates range from 41% to 88%.** Interpretation is confounded by inconsistent nomenclature and non-uniform VUR screening.
- It represents **transient non-pathological dilation. 21–56% resolves before birth.** "Mild" dilation resolves postnatally in **90–98%**, and much of it would now be **reclassified as normal** under UTD thresholds.
- **Registry timing:**
 - one registry: **>90% of SFU 1 and 2 resolved by 12 and 18 months** respectively;
 - **SFU registry: SFU 1 → 100% by about 26–30 months; SFU 2 → 95% by 48 months** (Fig. 34.15).
- **Caveat:** **some have undiagnosed VUR** (up to 31%). Parents and clinicians should know this.
- *Figure 34.15 — Resolution curves by initial SFU grade (Zee et al., 2017).* Grade 2 lags grade 1: 72% resolved at 2.5 years ($P < 0.05$).

Ureteropelvic Junction Obstruction Pattern of Dilation

- **UPJ obstruction is the commonest surgical cause of prenatal UTD.** Its incidence rises linearly with prenatal dilation. The discussion here is limited to **P2/P3 (SFU 3/4, moderate/severe).**
- **APD thresholds that predict surgery:**
 - **15 mm APD** → sensitivity about 75%, specificity about 80% for eventual pyeloplasty. Operated patients had a **mean APD of about 22 mm vs 12 mm** in those observed, and 15 mm was the optimal ROC cut-off.
 - **Second trimester 9.5 mm; third trimester 15 mm** (SFU registry data).
 - **Postnatal APD 20 mm** predicts **preservation of renal function** in early neonatal UTD.
- **Meta-analysis of the 15 mm threshold:** sensitivity about 0.8, specificity about 0.8, odds ratio about 13 for surgery. **Postnatal APD predicts surgery better than prenatal APD.**

- **APD varies with position, hydration and bladder filling.**
 - **APD falling from supine to prone** identified many who could be observed without surgery.
 - A later study found **no prone vs supine difference** in predicting pyeloplasty.
- **Severe UTD does not mean surgery:** in one series of severe UTD, only **about a fifth needed surgery**, and **about two-thirds resolved** on observation over about 2.5 years.
- **SFU grade predicts function and surgery:**
 - All SFU 3 kidneys had preserved function on scintigraphy.
 - Early intervention → mostly SFU 4; observation → mostly SFU 3.
- **Surgical treatment of UPJ obstruction is highly successful** (Chapter 53).

Recommendations for Ureteropelvic Junction Obstruction Pattern of Dilation

- **Most are followed without surgery.**
- **P2 or P3 → early neonatal US.**
- **PA may be considered for P3/SFU 4 (UTI risk).**
- **Lower tract imaging is recommended** to exclude VUR and confirm a **normal posterior urethra in males.**
- **Renal scintigraphy is discretionary**, but should be used for **persistent P3**, which correlates strongly with early surgery.
 - **Option 1:** baseline **well-tempered renogram at 1 month** for severe disease.
 - **Option 2:** **serial US** to determine the fate of the obstruction.

If **P2 and/or APD <20 mm and/or PT >3 mm → DRF >40% is likely** and surgery is unlikely to be needed. Neither approach is evidence based, so clinicians use the one they are comfortable with.

- **Consider surgery for:**
 - rising APD/UTD on serial US;
 - **DRF <40% with a delayed drainage curve;**
 - **falling DRF on serial renograms;**
 - **increased tracer retention on delayed upright MAG3 images;**
 - symptoms: **stone, pyelonephritis, haematuria.**

Vesicoureteral Reflux

- **VUR is found in up to 31% of screened prenatal UTD. Prenatal UTD grade correlates poorly with VUR severity.** Screening was historically justified by the high yield.
- **There are two clinical phenotypes of VUR** (Fig. 34.16: bilateral grade V VUR in P1 [SFU 2] UTD).
- *Early prospective multicentre series of prenatally detected VUR.*
 - **Mostly males, and mostly bilateral.**
 - **Ureteral dilation was uncommon, and a quarter had a normal first postnatal US.**
 - **About half of uncircumcised males developed UTI.**
 - High-grade units resolved **earlier than expected** (mean about 1 year in boys, 2 years in girls).
- **However, pooled data for the 2010 American Urological Association (AUA) VUR guideline showed no higher resolution rate for prenatally detected VUR** (overall resolution about 50%).
- **Whether prenatal VUR predicts UTI is disputed:**
 - **McMaster registry** (mostly high-grade UTD and high-grade VUR, two-thirds on PA): **VUR increased febrile UTI.** **SFU registry** (mostly low-grade UTD; VUR in about a fifth screened): **no association.** Both cohorts were mostly male; the difference reflects case mix (registry comparison above).

- **Practice patterns** (Fig. 34.17): for a 1-month-old with SFU 3 or 4, paediatric urologists recommend PA in **about a quarter to over half**, rising for **uncircumcised boys, girls and SFU 4**. They recommend VCUG in **about 85–90%** and follow-up RUS in all.

COMPARISON — PRENATALLY DETECTED VUR VS VUR PRESENTING WITH UTI

Feature	Detected by screening prenatal UTD	Presenting with febrile UTI
Sex	Male	Female
Grade / laterality	High grade, often bilateral	Lower grade
Renal damage	Congenital dysplasia (up to 40%) without UTI	Acquired scarring (less common)
UTI risk	Low if circumcised; high with intact foreskin	By definition presents with UTI
Resolution	Often early — attributed to high-pressure infant bladder dynamics that settle in the first years	Standard published rates

Recommendations for VUR

- Screen by UTI risk, not by the presence of dilation. Girls with high-grade UTD and uncircumcised boys → screen. Circumcised boys with low-grade UTD → no screening.
- Discuss natural history and the cost-benefit of prophylaxis with the family.

Megaureter/Ureterovesical Junction Obstruction

- **Primary congenital megaureter** is a less common cause of prenatal UTD. It is classified as **obstructed, non-obstructed, refluxing, or refluxing and obstructed**.
- **Distinguish from single-system ectopic ureter**, which is usually associated with **renal dysplasia** and inserts **distal to the bladder neck or into the vagina**.
- **Most resolve without surgery**:
 - half resolved by 7 years in one series; the rare late deterioration needed nephrectomy;
 - **about three-quarters resolved within 1.5–2 years** in others.
- **Predictors of surgery**:
 - ureteral diameter above about 13 mm (1.32–1.33 cm);
 - DRF <30%;
 - in prospective data, operated ureters averaged **17 mm vs 12 mm** in those observed;
 - about a quarter to a third came to surgery.

Recommendations for Megaureter

- **Shortly after birth: renal/bladder US and start PA; VCUG** to classify the megaureter.
- *British Association of Paediatric Urologists (BAPU) consensus (2014)*.
 - **Evaluation: prenatal ureter >7 mm after 30 weeks** is the threshold for postnatal RUS, PA and **VCUG in all**. Time the VCUG carefully if PUV is possible. Consider renal scintigraphy for function and UVJ obstruction.
 - **Surgery in a minority, for DRF <40%, febrile UTI or pain**.
 - **Under 1 year → temporising procedure**: cutaneous ureterostomy, cutaneous pyelostomy or refluxing reimplant.
 - **Definitive reimplantation is ideally deferred until after 1 year**.

- **Follow-up into adulthood:** late asymptomatic deterioration can occur.

Concept — Across UPJ, VUR and megaureter the same rule applies: **dilation alone is not an indication**. Intervention follows **loss of function, infection or progression**, and the grade of dilation mainly sets how closely to watch.

LUTO (PUV, PBS, URETHRAL STENOSIS/ATRESIA)

- **Prenatal signs of LUTO** (Fig. 34.18):
 - oligohydramnios;
 - bladder distension with **no cycling**;
 - ureteral and renal dilation, sometimes severe, with cystic change.
- **Imaging alone cannot separate PUV, urethral stenosis/atresia and PBS.**
- **A distended bladder compressing the thorax plus oligohydramnios → may need immediate pulmonary resuscitation. Highest risk: oligohydramnios before 27 weeks.**
- **Signs of a system under tension ("pop-offs"):**
 - **urinary ascites** (e.g. via a patent urachus);
 - **perirenal urinoma** (Fig. 34.19: urinoma and calyceal rupture on antegrade nephrostogram);
 - VUR;
 - renal dysplasia.
- **Pop-offs plus the "keyhole" sign** (dilated bladder and posterior urethra, Fig. 34.20) **favour PUV or urethral atresia**. However, **only fetal cystoscopy establishes the cause reliably**.
- **Window for intervention: 18–24 weeks**, after extensive counselling about benefit against the high complication rate.
- **At delivery the urology team should be ready to decompress the bladder.**
 - **PBS vs PUV on examination:** PBS has **absent abdominal wall musculature and bilateral impalpable testes**.
 - **Insert a 5 Fr feeding tube and confirm on KUB film.** It can curl in the **dilated posterior urethra** and still drain urine.
 - **Digital rectal pressure** helps guide the tube over a **high bladder neck**.
 - **Avoid balloon catheters:** the balloon can fall into the posterior urethra, obstruct or injure it, and impair drainage of the ureteric orifices.
 - **If a catheter cannot pass → bedside US-guided percutaneous suprapubic catheter**, using local anaesthetic and an angiocatheter by **Seldinger technique**.

Posterior Urethral Valves

- **Commonest cause of LUTO: 1.4–2.2 per 10,000 births.**
- **Diagnosed by examination and lower tract imaging.** VCUG shows a **dilated posterior urethra** (Fig. 34.21).
- **Prenatal diagnosis allows early neonatal treatment to preserve renal function** (Chapter 45).

Prune-Belly Syndrome/Megalourethra

- **PBS triad:** megacystis and upper tract dilation + **absent/deficient rectus abdominis musculature + bilateral impalpable testes** (Chapter 44).
- It may coexist with **urethral atresia** (→ oligohydramnios, with outcome similar to PUV) and with **congenital megalourethra**.

Urethral Stenosis/Atresia

- Presents with oligohydramnios, severe bilateral UTD and bladder distension.
- Postnatally a catheter will not pass. Cystoscopy shows a narrow penile urethra with a membrane at the membranous/prostatic urethra.
- Initial vesicostomy → then progressive augmented dilation of the anterior urethra (PADUA procedure).

Congenital Megacystis

- **Massive reflux form: laterally displaced ureteric orifices** → urine cycles between bladder and ureters → a large, thin-walled bladder. **Correcting the reflux restores bladder dynamics.**
- **Megacystis-microcolon-intestinal hypoperistalsis syndrome (MMIHS):**
 - the bladder is **distended but not obstructed**; VUR is rare;
 - prognosis was poor, but **survival has improved with parenteral nutrition and transplantation**;
 - most need **vesicostomy or intermittent catheterisation**;
 - **PA only for recurrent UTI.**

Recommendations for LUTO

- **Baseline renal/bladder US**, repeated after a period of decompression.
- **No immediate VCUG in a neonate with renal insufficiency who needs decompression:** replacing the catheter may be difficult.
- **Length of decompression is guided by the serum creatinine nadir.** Creatinine may **rise at first** from intravascular depletion or under-resuscitation.
- **Warn the neonatal team** about the **concentrating defect**, which is worsened by **post-obstructive diuresis**.
- **Oliguria or anuria after decompression** can occur because the **hypertrophied bladder wall resists drainage** from tortuous dilated upper tracts (Fig. 34.22: thick-walled bladder with distal ureteral dilation). **Antimuscarinics mitigate this.**
- **Repeat US as the baseline at maximal decompression → then VCUG** to guide surgery (Chapter 45).

COMPARISON — CAUSES OF BILATERAL UPPER AND LOWER TRACT DILATION

Condition	Oligohydramnios	Distinguishing feature	First neonatal step
PUV	Yes	Keyhole sign; dilated posterior urethra on VCUG	5 Fr feeding tube (confirm on KUB)
Urethral stenosis/atresia	Yes	Catheter will not pass ; membrane at membranous/prostatic urethra	Suprapubic drainage → vesicostomy → PADUA
PBS	If atresia coexists	Absent abdominal musculature + impalpable testes	Decompress; evaluate
Ureterocele (female)	Possible	Cystic bladder lesion	Decompress (puncture if obstructing)
Megacystis (massive reflux) / bilateral VUR	No	Laterally displaced orifices; thin bladder	Correct reflux
MMIHS	No	Non-obstructed distended bladder; microcolon; VUR rare	Vesicostomy or CIC

Concept — Oligohydramnios is the discriminator: bilateral dilation with normal fluid means urine is still leaving the fetus (reflux or megacystis), whereas bilateral dilation with low fluid means true outlet obstruction. The neonatal priority is then **safe decompression, not diagnosis**.

KIDNEY ANOMALIES

Multicystic Dysplastic Kidney

- **Most prenatal cystic disease needs no surgery** and spares overall renal function.
- **Prenatally, MCDK can be mistaken for P3 UTD** → review postnatal imaging critically. Whether MCDK is **very early severe obstruction** is debated.
- **Classic appearance: multiple non-communicating cysts of varying size with no large central cyst.** Reniform variants exist (Fig. 34.23).
- **Associated findings: VUR in about 20% and contralateral UPJ obstruction.**
- **Postnatal RUS confirms the diagnosis.**
- **VCUG and DMSA were historically routine but are now challenged:**
 - **VUR is found in under a fifth, is rarely surgical**, and unprophylaxed VUR has not caused febrile UTI;
 - **US diagnosis matched scintigraphy in 100%** when the bladder was normal on US.
- **Wilms tumour:** a systematic review of **over 1,000 children** found **no association between MCDK and Wilms tumour**. Routine nephrectomy has therefore been abandoned.
 - **Serial US confirms involution and compensatory contralateral hypertrophy.**
 - A cost analysis favoured **early nephrectomy**, but only by assuming long-term serial imaging.

Recommendations for Multicystic Dysplastic Kidney

- **Early postnatal renal/bladder US** to confirm.
- **No PA:** UTI risk is low.
- **VCUG and DMSA are recommended but not evidence based; omitting them is acceptable.** Long-term prospective data are needed.
- **Functional solitary kidney → monitor for hyperfiltration injury through puberty: blood pressure, proteinuria, and consider yearly US.** Serial imaging is not evidence based but ensures follow-up of a group at risk of adult renal disease.
- **The MCDK should involute. A persistent cystic kidney with a solid component → consider removal.**

Duplication Anomalies Ureterocele/Ectopic Ureter

- **Usually found prenatally because the upper-pole moiety is obstructed (Meyer–Weigert rule).**
- **Suspect duplication when ureteral dilation exceeds pelvic dilation** (Fig. 34.24). The dysplastic upper pole may be cystic.
- **A cystic mass in the bladder** is better seen on postnatal bladder US, which **separates ureterocele from ectopic ureter**. This is harder when the ureterocele opening is **ectopic in the bladder neck**.
- **In girls a ureterocele can cause LUTO with oligohydramnios** (Fig. 34.25: ball-valving ureterocele at cystoscopy).
- **Ureterocele disproportion:** a large ureterocele with minimal upper tract dilation (rare).
- **VUR is common and also follows Meyer–Weigert.** The upper pole inserts **distal and medial** (obstruction/ectopia); the lower pole inserts **cephalad and lateral** (short tunnel → reflux) (Fig. 34.26: ureterocele filling defect; "drooping lily" of lower-pole VUR).
- **Most duplication anomalies occur in females and are bilateral.**
- **Ureterocele vs ectopic ureter on US** (Fig. 34.27): a **ureterocele is a thin-walled cyst inside the bladder**. An **ectopic ureter may indent the bladder but shows two distinct walls**, with a muscle plane between ureter and bladder.

- **Initial management may be ureterocele puncture**, but this is not mandatory.
 - **Prenatal puncture** for outlet obstruction has been done successfully, but it is experimental and rarely needed.
 - **Traditionally, decompression was reserved for outlet obstruction or sepsis.**
 - Current practice increasingly uses **elective puncture as first-line therapy despite the reflux it creates.**
 - Some watch and wait if **diuretic renography shows no obstruction.**
- **Goal: preserve the lower-pole moiety.**
 - **Upper tract approach: nephron-sparing or ablative**, increasingly laparoscopic; **most ectopic ureters need nothing more.**
 - **Lower tract approach: common-sheath reimplantation** of both moieties, for those at most risk of UTI or with **high-grade lower-pole VUR.**
- **Presence and grade of VUR predict the need for lower tract surgery (Mayo series):**
 - **no VUR → upper-pole approach alone in all;**
 - **low-grade VUR → 40% needed further surgery for UTI;**
 - **high-grade VUR → ureterocele excision in 96%.**

Recommendations for Duplex Anomalies

- **Examine the introitus for a prolapsed ureterocele** (Fig. 34.28).
- **Early postnatal US** with attention to the bladder.
- **PA from birth.**
- **Characterise:** VUR, ureterocele size, bladder outlet obstruction.
- **Renal scintigraphy** for upper-pole function.
- **VUR grade and upper-moiety function dictate treatment:**
 - **Girl + ureterocele + high-grade lower-pole VUR + functioning upper pole → initial incision.** She is likely to need lower tract surgery anyway because of UTI risk.
 - **Boy + no VUR + dysplastic upper pole → upper-pole approach**, preserving or ablating.

Concept — *The Meyer–Weigert rule generates the whole duplex picture: **upper pole → caudal insertion → obstruction/ectopia; lower pole → cranial-lateral insertion → reflux.** Treatment follows from which moiety has the problem. **Lower-pole VUR pulls treatment to the bladder; its absence lets the upper tract be treated alone.***

EXSTROPHY (BLADDER/CLOACAL)

- **Prenatal diagnosis rarely leads to fetal intervention.** It allows counselling and referral to a **centre of excellence.**
- **Distinguishing bladder from cloacal exstrophy changes the natural history profoundly.**

Bladder Exstrophy

- **Prenatal hallmarks:**
 - **non-visualised bladder;**
 - **abnormal genitalia;**
 - **protruding infra-umbilical mass** (exstrophic bladder) below a **low-set umbilicus;**
 - **widened pubic diastasis.**

Detected prenatally in about 50% (Fig. 34.29: US and MRI of a male fetus showing a lower abdominal wall mass, urethral plate and penis).

- **Spine, kidneys and the rest of the fetal survey are usually normal**, unlike cloacal exstrophy. It is a **midline anterior wall defect** of bladder and genitalia without major organ system dysfunction.
- **Prenatal MRI** can confirm bladder exstrophy complex by surveying all organ systems.

Recommendations for Bladder Exstrophy

- **Not an emergency postnatally.** Examine thoroughly for **variants (e.g. bladder duplication)** that change the course. **Imaging includes RUS.**

Cloacal Exstrophy

- **Omphalocele–exstrophy–imperforate anus–spinal defects (OEIS) complex.** It is **extremely rare.**
- **Bladder exstrophy findings plus omphalocele and/or renal, vertebral or limb anomalies.**
- **Major and minor prenatal criteria** were derived from a small series combined with historical cases (Table 34.4). A **persistent cloacal membrane** was seen in a minority.
- **Early identification allows counselling at a centre of excellence.**

TABLE 34.4 CRITERIA FOR CLOACAL EXSTROPHY

Major criteria	Minor criteria
Non-visualisation of the urinary bladder	Lower extremity defects
Large midline infra-umbilical anterior wall defect or cystic anterior wall structure (persistent cloacal membrane)	Renal anomalies
Omphalocele	Ascites
Lumbosacral anomalies	Widened pubic arches
	Narrow thorax
	Hydrocephalus
	Single umbilical artery

Austin et al., 1998.

- **Purpose:** separate cloacal from bladder exstrophy prenatally.
- **Pattern:** **omphalocele and lumbosacral anomalies** are the major criteria that bladder exstrophy lacks. Non-visualised bladder is shared.
- **Trap:** a non-visualised bladder alone does not separate the two.

Recommendations for Cloacal Bladder Exstrophy

- **After delivery:** KUB, renal and spinal US, and consider MRI. The **extent of organ system involvement** determines surgery.

COMPARISON — BLADDER VS CLOACAL EXSTROPHY

Feature	Bladder exstrophy	Cloacal exstrophy (OEIS)
Bladder not seen, low umbilicus, pubic diastasis	Yes	Yes
Omphalocele	No	Yes
Spine / kidneys / limbs	Usually normal	Lumbosacral, renal, limb anomalies
Imperforate anus	No	Yes
Postnatal imaging	RUS	KUB, renal/spine US, ± MRI
Urgency	Not emergent	Dictated by organ systems involved

Cloacal Malformation/Urogenital Sinus/Congenital Adrenal Hyperplasia

- **Prenatal bladder plus vaginal distension → bladder outlet obstruction from vaginal trapping of urine.**
- **Differential diagnosis:**
 - **cloacal malformation** — expect **distal colonic obstruction visible prenatally**;
 - **urogenital sinus**;
 - **congenital adrenal hyperplasia (CAH)** — often a **family history (autosomal recessive)**.
- **Prenatal counselling is fundamental.**

Cloacal Malformation

- **Persistent cloaca: about 1 in 50,000 births; females only.** It results from **arrested development of the urorectal septum and urogenital sinus**, leaving rectum, vagina and urinary tract in one common channel.
- **Prenatal findings:**
 - **oligohydramnios, bilateral UTD and urinary ascites** (as in male LUTO);
 - **a large pelvic cystic mass**;
 - **calcified meconium** on MRI from rectal obstruction;
 - **dilated vagina/uterus**, which may be **bifid (uterus didelphys)** (Fig. 34.30: 30-week fetus with duplicated vagina, distended bladder and urinary ascites).
- **Urinary ascites arises by retrograde flow through the uterus and fallopian tubes into the peritoneum.** Oligohydramnios varies with the degree of gastrointestinal obstruction.
- **Polyhydramnios → consider an associated tracheo-oesophageal fistula.**
- **Delivery should be elective**, with neonatal, paediatric surgical and paediatric urological teams present.
- **Prenatal accuracy varies (rarity) → high suspicion; one series needed fetoscopy or fetal MRI to diagnose.**

UROGENITAL SINUS/CONGENITAL ADRENAL HYPERPLASIA

- **CAH is a rare autosomal recessive disorder** causing a **spectrum of genital ambiguity in females**.
- **Commonest form: 21-hydroxylase deficiency**; the gene is on the **short arm of chromosome 6**.
 - **Classic form: salt-wasting (75%) or simple virilising (25%).**
 - **Non-classic form: milder course.**
- **Incidence about 1 in 15,000–16,000 live births. Only 1 in 8 at-risk pregnancies produces an affected infant with ambiguity:** 1 in 4 is affected (recessive) × 1 in 2 is female, since **males are not ambiguous**.
- **Degree of urogenital sinus and virilisation varies**, so CAH can be confused with cloacal or urogenital sinus malformation.
- **Urogenital sinus malformation** can also trap urine in the vagina and cause LUTO, but **without gastrointestinal obstruction**. **Normal external genitalia favour urogenital sinus over CAH.**

COMPARISON — CLOACAL MALFORMATION VS UROGENITAL SINUS VS CAH

Feature	Cloacal malformation	Urogenital sinus malformation	CAH
Vaginal trapping of urine / LUTO	Yes	Yes	Possible
Gastrointestinal obstruction	Yes (calcified meconium)	No	No
External genitalia	Single perineal opening	Normal	Virilised / ambiguous

Feature	Cloacal malformation	Urogenital sinus malformation	CAH
Family history	No	No	Often (autosomal recessive)
Urgent neonatal issue	Decompression; bowel	Decompression	Salt-wasting crisis

Prenatal Treatment of Congenital Adrenal Hyperplasia

- **Prenatal identification and treatment of CAH is controversial.** Because only **1 in 8** at-risk pregnancies yields an affected girl, most fetuses exposed to prenatal treatment would not benefit.

Postnatal Treatment for Congenital Adrenal Hyperplasia

- **First concern: electrolyte imbalance.** Alongside stabilisation, confirm the diagnosis with **serum 17-hydroxyprogesterone** and **genetic testing**.
- **If vaginal retention of urine causes LUTO, bladder decompression is a priority.** Urinary and/or vaginal diversion may be needed (Chapters 59–60).

Outcomes for Fetal Myelomeningocele Closure

- **Prenatal alpha-fetoprotein (AFP) plus key US findings diagnose neural tube defects accurately.** Amniotic fluid **glial fibrillary acidic protein (AF-GFAP)** is a possible additional marker. Early diagnosis allows referral to a fetal surgery centre.
- *Multicentre randomised trial (2011), prenatal vs postnatal closure.*
 - **Prenatal closure before 26 weeks significantly reduced VP shunting** and appeared to **improve cognitive and motor function**.
 - **Main complications: preterm labour and uterine dehiscence.**
 - **Urology: no improvement in urodynamics, CIC need or need for urinary reconstruction.**

Upper Urinary Tract Obstruction

- **Urinary abdominal distension is usually obstructive** and anticipated prenatally.
- **The UPJ is the commonest site;** others are the **UVJ** and the **insertion of an ectopic ureter**.
- **Percutaneous drainage may be needed for respiratory compromise or feeding intolerance** (Fig. 34.31: massive upper-pole obstruction with ureterocele; mass effect relieved by ureterocele decompression).

KEY POINTS

- **Most prenatal UTD is transient** (resolution 41–88%).
- **Third-trimester APD of 15 mm** predicts the need for surgery.
- **VUR in up to 31% of prenatal UTD, commonest in males with bilateral high-grade disease.**
- **MCDK does not raise Wilms tumour risk** and can usually be observed.
- **Lower-pole VUR in duplex systems** influences how the obstructed upper pole is treated.
- **Megaureter usually resolves.** Operate for **recurrent UTI, reduced function or increasing UTD**, using **temporising diversion under 1 year**.
- **Fetal myelomeningocele closure** helps motor/cognitive outcome and VP shunt need but **not bladder outcomes or reconstruction need**.

Concept — In the female pelvic anomalies, the **vagina acts as a second bladder**: urine trapped there causes outlet obstruction, and urine escaping through the tubes causes ascites. **The companion findings pick the diagnosis**: gut obstruction means cloaca, virilisation means CAH, and normal genitalia mean urogenital sinus.

PERINATAL URGENCIES/EMERGENCIES

Neonatal Urologic Emergencies

- **Causes:** genitourinary anomalies, **birth trauma**, **vascular events**; **evaluate the whole newborn**.

Abdominal Distention

- **Causes:** **cystic lesion**, **solid mass**, **urinary ascites** (obstruction or extravasation); examine the whole abdomen.

Lower Urinary Tract Obstruction

Urinary Ascites

- **Urine causes up to 30% of neonatal ascites.** Features:
 - respiratory distress;
 - oliguria/anuria;
 - hyponatraemia and hyperkalaemia;
 - raised creatinine.
- **PUV is the commonest cause of urinary ascites.** Others: **bladder rupture**, **cloacal malformation**, **persistent urogenital sinus**, **urethral atresia**.
- **Suspected bladder perforation:**
 - **catheterise**, and follow urine output and electrolytes;
 - **confirm by biochemical analysis of peritoneal fluid plus VCUG**;
 - often unexplained if there is no outlet obstruction;
 - **most spontaneous perforations heal conservatively**;
 - **iatrogenic rupture from umbilical catheters** needs prompt recognition and repair.
- **Urogenital sinus or cloaca:** urine pools in the vagina, flows retrograde through the uterus and **fallopian tubes into the peritoneum**. VCUG demonstrates it.
- **Percutaneous drainage** if the distension compromises breathing.

Urinary Retention (>24 Hours After Birth)

- **Mostly males with LUTO:** PUV, urethral atresia, rarely ureterocele. **In females, suspect a urogenital sinus anomaly with vaginal distension**; ureterocele can also cause retention.
- **Relieve the obstruction** as well as working up the cause.
- **Catheterise.**
 - **PUV with a high bladder neck** → **finger in the rectum** to deflect the catheter over the neck.
 - **Females:** aim for the bladder, but **draining the vagina may suffice** and may allow voiding.
- **If catheterisation fails** → **percutaneous drainage by Seldinger technique: 16-gauge angiocatheter** → **3.5 Fr feeding tube**; 14-gauge → 5 Fr feeding tube.
- **Some urogenital sinus anomalies:** intermittent vaginal drainage does not drain the bladder → **vesicostomy and/or vaginostomy** until definitive repair (Fig. 34.32: persistent bladder and vaginal dilation despite catheterisation, relieved by vaginostomy/vesicostomy, then normal cycling after vaginoplasty).

Solid or Cystic Renal Masses

- **Four categories of abdominal mass:** **renal cystic**, **pelvic cystic**, **solid**, **non-genitourinary**.
- **Renal cystic:** MCDK, ARPKD, autosomal dominant polycystic kidney disease (ADPKD), cystic nephroma.
 - **MCDK = multiloculated mass with little or no parenchyma.**

- **Image both sides** (contralateral VUR/UPJ obstruction).
- **Solid renal lesions are uncommon in newborns.**
 - **Commonest solid renal tumour under 6 months = congenital mesoblastic nephroma (CMN)**, associated with **hypertension** and **maternal polyhydramnios**.
 - **Wilms tumour** usually presents at 1–5 years but can be present at birth.
 - **Others:** ossifying renal tumour, rhabdoid tumour, clear cell sarcoma, neuroblastoma.
 - They present with **mass and haematuria**; paraneoplastic **hypertension and hypercalcaemia** occur.
- **Postnatal US plus family history** (ADPKD/ARPKD) separates cystic from solid lesions. Sonographic findings direct further work-up.
- **Adrenal haemorrhage:**
 - more common on the **right; bilateral in 10–15%**;
 - causes: **birth trauma, hypoxia, sepsis, coagulopathy**;
 - **echogenic adrenal mass with no Doppler flow**;
 - ranges from an incidental finding to life-threatening bleeding, presenting with **anaemia, jaundice, hypotension, mass**;
 - **scrotal haematoma** via a patent processus vaginalis can mimic an acute scrotum;
 - **main differential is neuroblastoma** (MRI if needed); **serial US every 1–2 weeks** should show shrinkage.
- **Pelvic cystic lesions:**
 - **distended bladder or urinary ascites**;
 - **hydrometrocolpos** — vagina distended with cervical secretions from **vaginal atresia, transverse septum or imperforate hymen**; it can be mistaken for a distended bladder;
 - **urogenital sinus/cloaca** — vagina distended with urine;
 - **ovarian cysts** — present in **up to 30% of female neonates**; **anechoic unilocular pelvic mass**; most resolve within 6 months.

COMPARISON — RESOLVING ADRENAL HAEMORRHAGE VS NEONATAL NEUROBLASTOMA

Feature	Adrenal haemorrhage	Neuroblastoma
Doppler	No flow	Vascular solid mass
Serial US (1–2-weekly)	Shrinks	Persists or grows
Calcification	Peripheral eggshell (when resolving)	Stippled
Context	Birth trauma, hypoxia, sepsis, coagulopathy	—

Anorectal Malformation

- **Genitourinary and spinal anomalies occur in up to 75% of children with anorectal malformation** → screen all.
- **Commonest:** VUR, renal dysplasia, renal agenesis, cryptorchidism.
- **Initial work-up:**
 - inspect the **perineum and back**;
 - renal US;
 - **spinal US in the newborn period** for tethered cord.
- **VUR assessment with videourodynamics at 6 weeks; urinary prophylaxis until then.**

Ambiguous Genitalia

- **Urgent, anxiety-laden consultation:** involve a multidisciplinary **disorders of sex development (DSD)** team (endocrinologist, psychiatrist, geneticist); aim for **accurate diagnosis, then gender assignment and counselling — defer assignment until diagnosis.**
- **Commonest cause of ambiguity at birth: CAH** (up to 1 in 16,000). **46,XX DSD is the commonest DSD category**, from fetal androgen exposure (enzyme deficiency or maternal androgens).
 - **No palpable gonads.**
 - **US shows a uterus and ovaries.**
- **First priority:** exclude classic salt-wasting CAH. Loss of aldosterone → **hyperkalaemia and hyponatraemia** → dehydration and arrhythmias.
- **Tests:** karyotype, electrolytes, 17-hydroxyprogesterone, testosterone, dihydrotestosterone.
- **Less common:** 46,XY DSD, 46,XX testicular DSD, sex chromosome DSD, ovotesticular DSD. These may need gonadal biopsy (Chapter 59).

Interlabial Masses in a Female

- **Four diagnoses:** imperforate hymen, prolapsed urethra, prolapsed ureterocele, paraurethral cyst.
- **Diagnosis is clinical;** a **cotton-tipped applicator** defines the anatomy.

COMPARISON — NEONATAL INTERLABIAL MASSES

Mass	Position	Appearance	Management
Imperforate hymen (hydrometrocolpos)	Midline	White bulging membrane at introitus ± UTD; fluid behind bladder on US	Hymenal incision → milky secretions drain
Urethral prolapse	Midline	Circumferential oedematous, ecchymotic collar at meatus; uncommon in newborns; commoner in Black girls	Conservative: warm compresses, topical emollients, avoid straining/coughing
Prolapsed ureterocele	Off midline; intact hymen posterior	Cystic structure emerging from the urethral meatus	Confirm with US and VCUG
Paraurethral (parameatal) cyst	Off midline; intact hymen posterior	Opalescent ovoid cyst , may displace meatus/deflect stream	Most resolve spontaneously

- **Pattern:** midline → hymen or urethra; off-midline with an intact hymen behind → ureterocele or paraurethral cyst.

Oliguria/Anuria

- **Most neonates void within 24 hours;** later voiding is **generally benign.**
- **Gentle suprapubic palpation** may trigger voiding; **US** can show a distended bladder.
- **After circumcision:** 75% of breastfed and 100% of formula-fed boys void within 8 hours.

Gross Hematuria

- **Wide differential** (Table 34.5). **First confirm urinary origin with urinalysis and microscopy.**
- **Mimics:**
 - **maternal oestrogen withdrawal bleeding** in girls (self-limiting);
 - dermatitis;
 - post-circumcision bleeding;

- **red urate crystals.**
- **Initial imaging: abdominal US with Doppler** for the vessels.
- **Haematuria plus hypertension = more severe disease:** suspect thrombosis, polycystic kidney disease or cortical necrosis.

TABLE 34.5 CAUSES OF NEONATAL GROSS HEMATURIA

Category	Causes
Vascular	Renal vein thrombosis; renal artery thrombosis; bleeding disorder (vitamin K deficiency, thrombocytopenia)
Urinary tract obstruction	UPJ obstruction; posterior urethral valves
Anatomical	ARPKD; ADPKD; medullary sponge kidney; arteriovenous malformation
Trauma	Suprapubic aspiration / traumatic catheterisation
Tumours	Wilms tumour; mesoblastic nephroma
Renal (parenchymal)	Renal cortical/papillary necrosis; glomerulonephritis

- **Pattern: vascular causes lead** in the neonate.
- **Trap: bloody diaper ≠ haematuria** until urinalysis confirms it.

COMPARISON — NEONATAL RENAL VASCULAR CAUSES OF HAEMATURIA

Feature	Renal vein thrombosis (RVT)	Renal artery thrombosis (RAT)	Renal cortical necrosis
Frequency	Commonest thrombotic event; up to 20% of neonatal gross haematuria	Less common	Less common
Cause / risk factors	Maternal diabetes, prematurity, umbilical venous catheter, polycythaemia; up to 50% have a prothrombotic disorder	Umbilical artery catheter (chief cause)	Any cause of renal hypoperfusion: hypotension, anoxia, haemorrhage, necrotising enterocolitis, sepsis, abruption, twin–twin transfusion
Presentation	Triad: gross haematuria + flank mass + thrombocytopenia	Hypertension, congestive heart failure	Renal failure
Imaging	Doppler US	Renal scan: cortical perfusion defects	RUS: lost corticomedullary differentiation, hypoechoic cortical rim
Treatment	Aggressive hydration, expectant management; consider anticoagulation; nephrology; haematology work-up	Remove umbilical artery catheter ± thrombolysis	Supportive; lifelong risk of renal failure

Scrotal Mass in the Neonate

- **First establish acuity** (staff history, palpation) (Table 34.6).

TABLE 34.6 CAUSES OF SCROTAL MASSES IN THE NEONATE

Group	Causes
Vascular/torsion	Extravaginal torsion; appendage torsion
Processus vaginalis-related	Incarcerated hernia; hydrocele; meconium (meconium peritonitis); haematoma from adrenal haemorrhage; tumour seeding (renal or adrenal tumours)

Group	Causes
Intrascrotal	Epididymal cyst; testis tumours; trauma

- **Examination aids:**
 - **transillumination** → hydrocele;
 - **gentle palpation** → reducible hernia or communicating hydrocele;
 - **abdominal radiograph** → scrotal bowel gas (hernia) or **stippled calcification** (meconium peritonitis).
- **Wilms tumour or neuroblastoma can seed the testis** via the processus. **Scrotal haematoma can mimic torsion.**
- **Neonatal testis tumours are rare:** yolk sac tumour, teratoma, **juvenile granulosa cell tumour**, gonadal stromal tumours. If malignancy is suspected, perform an **abdominal US**.

Testicular Torsion

- **Neonatal torsion is usually extravaginal** (tunica vaginalis and cord twist together).
- **Prenatal (in-utero) vs postnatal (first 30 days)** torsion must be distinguished.
 - **Salvage after prenatal torsion is nearly nil;** after postnatal extravaginal torsion it is **up to 44%.**
 - Survey: **43% of paediatric urologists treat prenatal torsion as an emergency vs 93% who would immediately explore postnatal torsion.**
- **Both types:** heterogeneous echotexture, ± hydrocele, and **no Doppler flow.**
- **Nonsalvageable prenatal torsion → urgent contralateral orchidopexy** to prevent **asynchronous contralateral torsion.** Consensus on urgency, or on whether to explore at all, is lacking. The risk of extravaginal torsion is thought to last only **the first few weeks**, so act before that window closes.
- **Postnatal torsion → manage as acute torsion later in life**, with timing adapted to presentation. Parents must know that **salvage is low.** **The main rationale for exploring is preventing contralateral torsion.** **Spinal anaesthesia** avoids neonatal inhalational agents.

COMPARISON — PRENATAL VS POSTNATAL NEONATAL TESTICULAR TORSION

Feature	Prenatal (in utero)	Postnatal (first 30 days)
Scrotum at birth	Already enlarged, discoloured	Normal at birth → new erythema/swelling
Testis	Non-tender, hard, fixed to scrotal wall	Tender
US	Hyperechoic, avascular; heterogeneous	Heterogeneous; absent flow
Salvage	Nearly nil	Up to 44%
Surgery	Urgent (not emergent) contralateral orchidopexy ± exploration	Immediate exploration + contralateral fixation

KEY POINTS

- **Urological causes of neonatal abdominal distension:** urinary ascites, upper or lower tract obstruction, cystic or solid renal mass, and rarely a retroperitoneal (adrenal) mass.
- **Urinary retention** is caused by LUTO: **PUV in males, urogenital sinus or cloaca in females.**
- **Neonatal renal tumours are uncommon;** the commonest is **mesoblastic nephroma.**
- **Ambiguous genitalia → check electrolytes for salt-wasting CAH;** defer gender assignment.
- **Neonatal gross haematuria** is uncommon; investigate with **urinalysis and Doppler abdominal US.**

- **The acuity of neonatal torsion** dictates the need for emergency exploration.

Concept — Neonatal emergencies are dominated by **what the newborn brought with it** (obstruction, ascites, anomalies) and by **what birth and neonatal care did to it** (trauma, hypoxia, umbilical catheters). Timing, prenatal versus postnatal, determines whether an organ can still be saved. **Torsion is the model case:** in-utero torsion is operated on for the other testis, and postnatal torsion is explored to save both.